

Problem 1

Let us, denote the patient arrival process by N_t , and the events:

$$\begin{aligned}\{\text{patient } i \text{ is sick with coronavirus}\} &=: C_i, \\ \{\text{patient } i \text{ is asymptomatic}\} &=: A_i, \\ A_i^c &=: S_i.\end{aligned}$$

From the problem statement we know that for all $i \geq 1$:

$$\begin{aligned}\mathbb{P}(C_i) &= 0.8, \\ \mathbb{P}(A_i|C_i) &= 0.25, \\ S_i \cap C_i &= S_i.\end{aligned}$$

a. Let us first calculate the probability that the first patient is exhibiting symptoms:

$$\mathbb{P}(S_1) = \mathbb{P}(S_1 \cap C_1) = \mathbb{P}(S_1|C_1)\mathbb{P}(C_1) = (1 - 0.25) \times 0.8 = 0.6.$$

By thinning, the number of symptomatic patient arrivals by time t , N_t^S , will be a Poisson process with a rate $\mathbb{P}(S_1) \times 20 = 12$. Thus, the first arrival time of a symptomatic patient, T_1^S , will be distributed exponentially with a parameter 12, $T_1^S \sim \text{Exp}(12)$. Therefore:

$$\mathbb{E}[T_1^S] = \frac{1}{12}.$$

b. The number of infected patients by time t , N_t^C , is composed of the number of asymptomatic patients that are infected $N_t^{C,A}$, and the number of symptomatic patients $N_t^{C,S}$. Since $\mathbb{P}(A_i \cap C_i) = \mathbb{P}(A_i|C_i)\mathbb{P}(C_i) = 0.2$ and $\mathbb{P}(S_i \cap C_i) = 0.6$, by thinning $N_t^{C,A}$ and $N_t^{C,S}$ are Poisson process with the respective rates 4 and 12. By superpositioning N_t^C is a Poisson process with rate 16, and the label of each arrival is independent from the process itself. The event of interest can be expressed as:

$$\{\text{The first arrival of } N_t^C \text{ is asymptomatic}\},$$

and thus

$$\mathbb{P}(\{\text{The first arrival of } N_t^C \text{ is asymptomatic}\}) = \frac{4}{16} = 0.25.$$

c. Given that the first 3 patients are asymptomatic, it is more convenient to compute the probability that none of the patients are sick:

$$\begin{aligned}\mathbb{P}(C_1^c \cap C_2^c \cap C_3^c | A_1 \cap A_2 \cap A_3) &= \frac{\mathbb{P}(C_1^c \cap A_1 \cap C_2^c \cap A_2 \cap C_3^c \cap A_3)}{\mathbb{P}(A_1 \cap A_2 \cap A_3)} \\ \text{since } C_i, A_i \perp\!\!\!\perp C_j, A_j, \text{ for all } i \neq j: &= \frac{\prod_{i=1}^3 \mathbb{P}(C_i^c \cap A_i)}{\prod_{i=1}^3 \mathbb{P}(A_i)} = \left(\frac{\mathbb{P}(C_1^c \cap A_1)}{\mathbb{P}(A_1)} \right)^3 \\ \text{since } A_1 \cap C_1^c &= C_1^c: &= \left(\frac{\mathbb{P}(C_1^c)}{\mathbb{P}(A_1)} \right)^3 = \left(\frac{1 - \mathbb{P}(C_1)}{1 - \mathbb{P}(S_1)} \right)^3 = \left(\frac{0.2}{0.4} \right)^3 \\ &= 0.125.\end{aligned}$$

Thus, $\mathbb{P}(C_1 \cup C_2 \cup C_3 | A_1, A_2, A_3) = 1 - 0.125 = 0.875$.

Grading Comments:

Many students made a minor mistake that the expectation of exponential is not the inverse of its rate. -2 for wrong answers from each item. If solutions were not fully developed with wrong answer, -3 were taken off from each item. Most students got item b) correctly whereas a majority of points were taken off from item c). In this case, students wrongly computed the probability that none of them is sick which is in fact $1/2$, and taking that to the power of 3rd could have solved the problem easily. Students with wrong computations had -2 points off.

Problem 2

Let us denote the number of packages in the shipment by $N \sim \text{Poisson}(4)$.

a. The event of interest can be characterized as $\{N \geq 3\}$, thus:

$$\begin{aligned}\mathbb{P}(N \geq 3) &= 1 - \mathbb{P}(N < 3) = 1 - \mathbb{P}(N = 0) - \mathbb{P}(N = 1) - \mathbb{P}(N = 2) \\ &= 1 - e^{-4} - 4e^{-4} - \frac{1}{2}4^2e^{-4} = 1 - 13e^{-4} \approx 0.76.\end{aligned}$$

b. Let $A_i := \{\text{ith customer buys a package}\}$, and the demand for the wipes be $D = \sum_{i=1}^3 \mathbf{1}_{A_i} \sim \text{Binom}(3, 0.5)$. Then the total number of packages of wipes the store sells can be represented as $N \wedge D$. Using the law of total expectation we get:

$$\begin{aligned}\mathbb{E}[N \wedge D] &= \sum_{d=0}^3 \mathbb{E}[N \wedge d | D = d] \mathbb{P}(D = d) \\ N \perp\!\!\!\perp D : &= \sum_{d=1}^3 \mathbb{E}[N \wedge d] \mathbb{P}(D = d).\end{aligned}$$

Now, let us calculate each summand:

$$\begin{aligned}\mathbb{E}[N \wedge d] &= \sum_{n=0}^{\infty} (n \wedge d) \mathbb{P}(N = n) = \sum_{n=0}^{d-1} n \mathbb{P}(N = n) + \sum_{n=d}^{\infty} d \mathbb{P}(N = n) = \\ &= d - \sum_{n=0}^{d-1} (d - n) \mathbb{P}(N = n) = d - \sum_{n=0}^{d-1} (d - n) \frac{4^n e^{-4}}{n!}.\end{aligned}$$

In particular for $d \in \{1, 2, 3\}$ we get:

$$\begin{aligned}\mathbb{E}[N \wedge 1] &= 1 - e^{-4}, \\ \mathbb{E}[N \wedge 2] &= 2 - 2e^{-4} - 4e^{-4} = 2 - 6e^{-4}, \\ \mathbb{E}[N \wedge 3] &= 3 - 3e^{-4} - 8e^{-4} - 0.5 \times 16e^{-4} = 3 - 19e^{-4}.\end{aligned}$$

Thus, since $D \sim \text{Binom}(3, 0.5)$ we get:

$$\begin{aligned}\mathbb{E}[N \wedge D] &= \frac{1}{2^3} \left(3(1 - e^{-4}) + 3(2 - 6e^{-4}) + (3 - 19e^{-4}) \right) \\ &= \frac{1}{8} (12 - 40e^{-4}) \approx 1.41.\end{aligned}$$

Grading Comments:

- a. For question (a) the majority of students got right (of course). Yet there are still many of them who got wrong. A common mistake was just either forgetting one of the terms (e.g. $P(N = 0)$) or adding another one (e.g. $P(N = 3)$). But a non-negligible number of students also made more serious mistakes like confounding pmf and cdf, and confounding addition with multiplication when developing the probability of disjoint events (e.g. some wrote $P(A \cup B) = P(A) \cdot P(B)$)...
- b. For question (b) most students couldn't really derive explicitly the quantity that was meant to be computed (i.e. $E[\min(D, N)]$) in one way or another. Hence, they were left somehow guessing what expectation they should compute, and the most common mistake was for them to simply give as answer the expectation of a $\text{Bin}(3, 0.5)$ (that is just $3/2$), representing the three customers buying or not. Sometimes they multiplied this answer by the probability of having $N \geq 3$. Another common mistake was to wrongly use the independence between D and N in the following way: in this case, they generally started to decompose the expectation by conditioning on the event $\{D \leq N\}$, and its negation, and then were saying that, by independence, we have for instance $E[D|D \leq N] = E[D]$. Some other students tried also to use wrongly thinning theorem. Finally, many students actually succeeded in setting up the problem properly, but made some calculus errors along the way.

Problem 3

Let the interval $[0, 24]$ represent the Trader Joe's parking lot, and without loss of generality let's assume that the entrance is located at 0. Let X_i be the distance of the i -th customer's car from the entrance, which yields that $X_i \sim \text{Uniform}(0, 24)$.

- a. The event of interest can be represented as $\{X_1 \in [0, 6]\}$, and thus

$$\mathbb{P}(X_1 \in [0, 6]) = \frac{6}{24} = 0.25.$$

Since $X_1 \sim \text{Uniform}(0, 24)$, the expected distance will be given by:

$$\mathbb{E}[X_1] = \frac{24}{2} = 12.$$

b. The mean square distance will be given by:

$$\mathbb{E}[(X_1 - X_2)^2] = \mathbb{E}[X_1^2] + \mathbb{E}[X_2^2] - 2\mathbb{E}[X_1 X_2]$$

$$\text{since } X_1, X_2 \text{ are iid: } = 2\mathbb{E}[X_1^2] - 2\mathbb{E}[X_1]^2 = 2\text{Var}(X_1) = 2\frac{24^2}{12} = 96.$$

c. The event of interest can be represented as $\{|X_1 - X_2| > 6\}$. Note that since X_1 and X_2 i.i.d., then $X_1 - X_2$ and $X_2 - X_1$ are identically distributed, and thus:

$$\mathbb{P}(|X_1 - X_2| > 6) = \mathbb{P}(X_1 - X_2 > 6 \cup X_2 - X_1 > 6)$$

$$\text{disjoint events} = \mathbb{P}(X_1 - X_2 > 6) + \mathbb{P}(X_2 - X_1 > 6)$$

$$X_1 - X_2, X_2 - X_1 \text{ i.d.} = 2\mathbb{P}(X_1 - X_2 > 6).$$

Applying the law of total probability:

$$\begin{aligned} \mathbb{P}(X_1 - X_2 > 6) &= \int_{\mathbb{R}} \mathbb{P}(X_1 - X_2 > 6 | X_2 = y) \mathbb{P}(X_2 \in dy) \\ &= \int_{\mathbb{R}} \mathbb{P}(X_1 - y > 6 | X_2 = y) \mathbb{P}(X_2 \in dy) \\ X_1 \perp\!\!\!\perp X_2 : &= \int_{\mathbb{R}} \mathbb{P}(X_1 - y > 6) \mathbb{P}(X_2 \in dy). \end{aligned}$$

Now let us calculate the integrand:

$$\begin{aligned} \mathbb{P}(X_1 - y > 6) &= \frac{1}{24} \int_{\mathbb{R}} \mathbf{1}_{\{x-y>6\}} \mathbf{1}_{\{x \in [0,24]\}} dx \\ &= \frac{1}{24} \int_{\mathbb{R}} \mathbf{1}_{\{x-y>6\}} \mathbf{1}_{\{x \in [0,24]\}} \mathbf{1}_{\{y \in [0,18]\}} dy \\ &= \frac{18-y}{24} \mathbf{1}_{\{y \in [0,18]\}}. \end{aligned}$$

Thus,

$$\mathbb{P}(X_1 - X_2 > 6) = \frac{1}{24} \int_{\mathbb{R}} \frac{18-y}{24} \mathbf{1}_{\{y \in [0,18]\}} \mathbf{1}_{\{y \in [0,24]\}} dy = \frac{1}{24^2} \int_0^{18} 18-y dy = \frac{18^2}{2 \times 24^2} = \frac{9}{32}.$$

Finally this gives us $\mathbb{P}(|X_1 - X_2| > 6) = \frac{9}{16} = 0.5625$.

Grading Comments:

- Some students misinterpreted the uniform distribution here as a discrete random variable.
- Some students were confused by the definition of the mean square distance. Let X_1 and X_2 denotes the spots of the first two customers, then "mean" here means expectation, "distance" means $|X_1 - X_2|$, so "mean square distance" means $\mathbb{E}[|X_1 - X_2|^2]$.
- Many students correctly applied the idea of tower rule/law of total expectation while ignoring the conditioning/having the wrong integral range/missing the indicator functions. Note once we fix X_1 and calculate the integral with respect to X_2 , things will behave differently depending on the value of X_1 .